Manager Report | Walmart Store Sales Forecasting dataset

EVALUATING MACHINE LEARNING MODELS FOR WEEKLY RETAIL SALES FORECASTING

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OVER ALL RECOMENDATION

Use XGBoost as the leading candidate for a controlled pilot, alongside the Naive baseline. Across six chronological test folds, XGBoost had the lowest mean MAE (1,704) and lowest mean MASE (0.877). Linear Regression had the lowest mean RMSE (4,161), suggesting it performed better on this metric that penalizes large errors. Do not automate ordering solely from this experiment; validate forecasts in a live pilot and review high-error departments.

1. Purpose

Determine whether a one-week sales prediction is trustworthy enough to guide decisions about purchases, staffing, inventory, and replenishment.

2. Key results at a glance:

Model	MAE Mean	MAE Std	RMSE Mean	RMSE Std	MASE Mean	MASE Std
Linear Regression	1,792.09	582.23	**4,160.69**	2,237.53	0.911	0.416
Naive	2,213.96	1,262.26	5,866.60	4,981.04	1.150	0.827
XGBoost	**1,703.68**	750.76	4,249.13	2,903.86	**0.877**	0.512

Inter petition :

MAE is the average absolute prediction error in dollars.
RMSE penalizes larger errors more heavily.
MASE compares a model's MAE with the Naive baseline; values below 1 mean better performance than the baseline.

3. Business Problem and Data

A business decession maker can plan inventory, staffing levels, replenishment, and distribution with the use of retail sales estimates. Over-forecasting can result in waste and higher holding costs, while under-forecasting can cause stockouts and lost revenues. In this project, the question is: Can those decisions be supported by a forecast that is one week ahead of time?

Here used dataset contains weekly sales by Store and Department, joined to calendar/economic features and store metadata. The raw training file contains 421,570 rows, 45 stores, and 81 departments. Its date range is 2010-02-05 to 2012-10-26. The analysis forecasts Weekly_Sales for each Store–Department series one week ahead.

3.1 Data sources and preparation

- The project uses the supplied Walmart Store Sales Forecasting files: train.csv, features.csv, and stores.csv.
- The training data contains weekly sales by Store and Department, alongside date and holiday information.
- Features and store metadata were merged using Store and Date where appropriate.
- Missing Markdown values were set to zero.
- Calendar variables (year, month, week, quarter) and lag/rolling features were created within each Store–Department series.

- Lag features included previous-week, two-week, four-week, and 52-week sales.
- Four-week rolling mean and standard deviation were shifted by one week before calculation to avoid using the current week's target.
- Rows without required lag or rolling values were removed.

Important scope note: The sales trend graphic aggregates observations across the included stores/departments by date; the error statistics are per Store-Department weekly observation, not the error for a single store or for all Walmart sales.

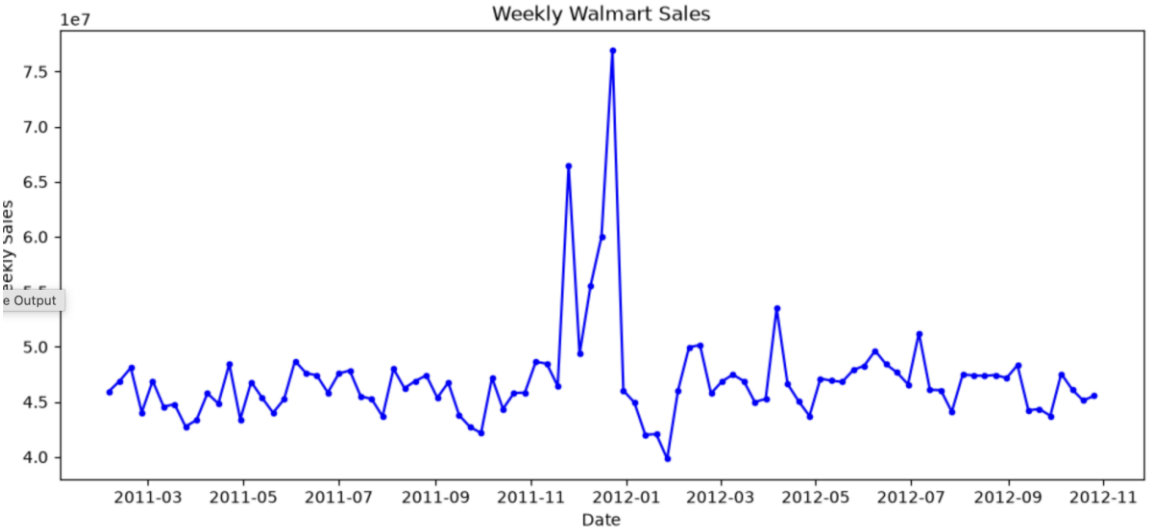


Figure 1. Total weekly sales across Store–Department observations in the training data. The aggregate chart shows changes in sales level over time; it does not by itself demonstrate forecast accuracy

4. Testing Design

Rather of using a random train/test split, the data were assessed using a chronological walk-forward method. There was a one-week interval between the completion of training and the commencement of the test, and six different test folds with ten weekly dates each were used. Models were only trained on days before to the test window and assessed on subsequent dates for each fold. For every model, the same dates, target, horizon, and metrics were applied.

Lag features for a test week are based on previously observed sales, and the evaluation is a rolling one-step-ahead test.

When the most recent completed week's actual sales are available before the subsequent projection is released, this is similar to weekly operational forecasting. A 10-week recursive prediction generated all at once is not evaluated by it.

Models compared

Naive baseline: forecast this week's sales using the previous week's sales for the same Store–Department series (lag1). This is a simple benchmark that a more complex model should beat.

Linear Regression: a simple model using calendar variables and sales lags/rolling summaries.

XGBoost: a boosted-tree model intended to capture nonlinear relationships among the same engineered features. XGBoost results are taken from the submitted notebook's completed runs for folds 1–6; the notebook's results CSV duplicated those six folds a second time, so duplicate rows were excluded from this report.

Metrics

MAE measures the typical absolute error in sales dollars. RMSE gives extra weight to large misses. MASE is MAE divided by the Naive model's MAE for the same fold; MASE = 1 matches the baseline and MASE below 1 beats it. Results are summarized as the mean and sample standard deviation across the six folds.

5. Results

While Linear Regression produced the lowest average RMSE, XGBoost produced the lowest average MAE and MASE.This indicates that while Linear Regression's mistakes were somewhat better under a criteria that penalizes huge misses, XGBoost was superior for average absolute error. On average, both trained models performed better than the Naive baseline, despite significant fold-to-fold variation.

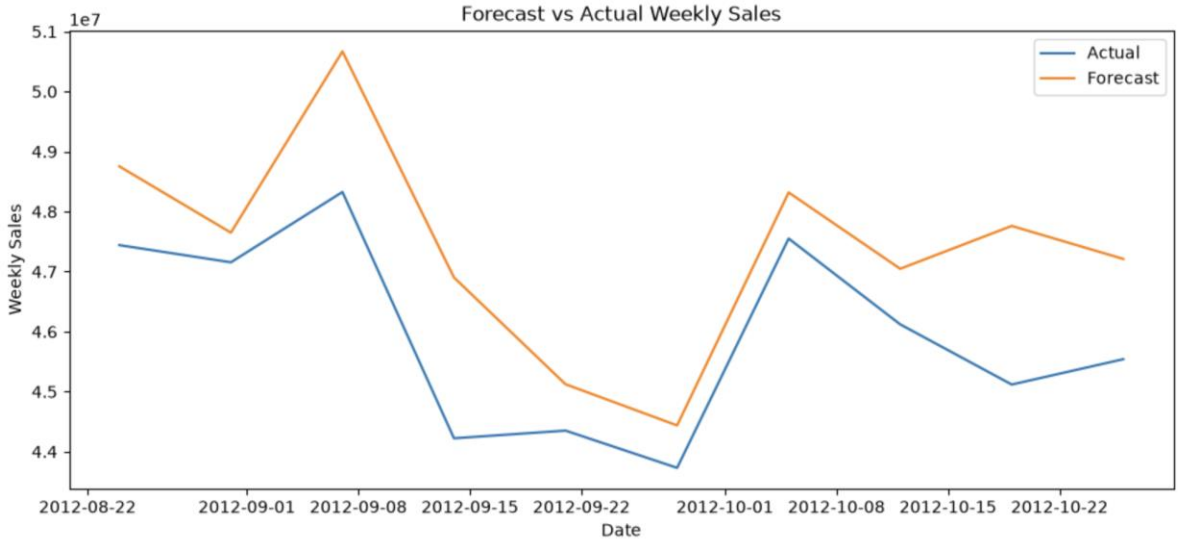


Figure 2. Forecast vs Actual Weekly Sales (original graph from the project notebook) The graph

shows the aggregate actual and Linear Regression forecast sales across the final test window. Aggregate results can hide errors in individual Store–Department series.

6. Performance Across Test Folds

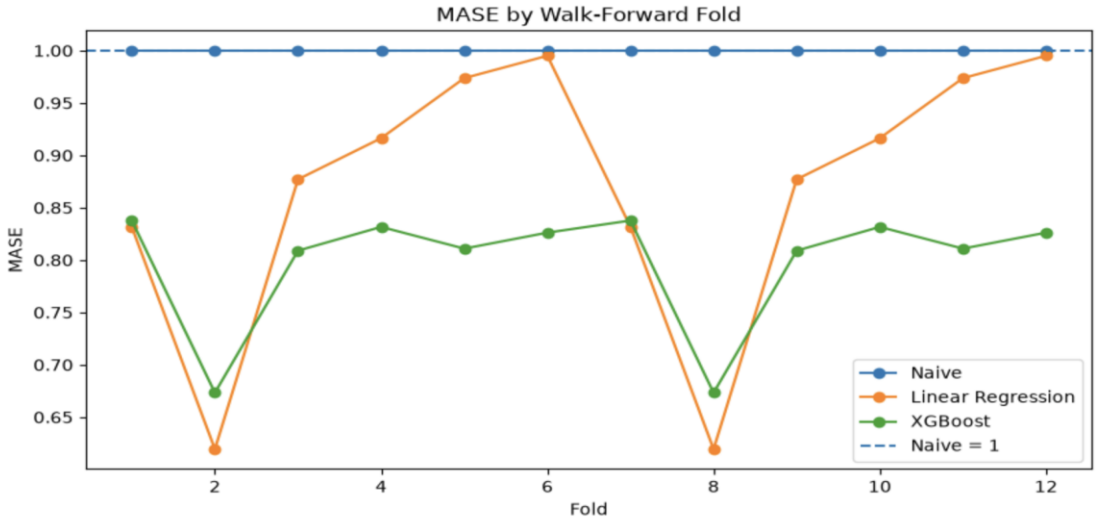


Figure 3. MASE by fold. Values below the dashed line (1.0) outperform the Naive baseline. Variation across folds indicates that performance depends on the time period and sales conditions.

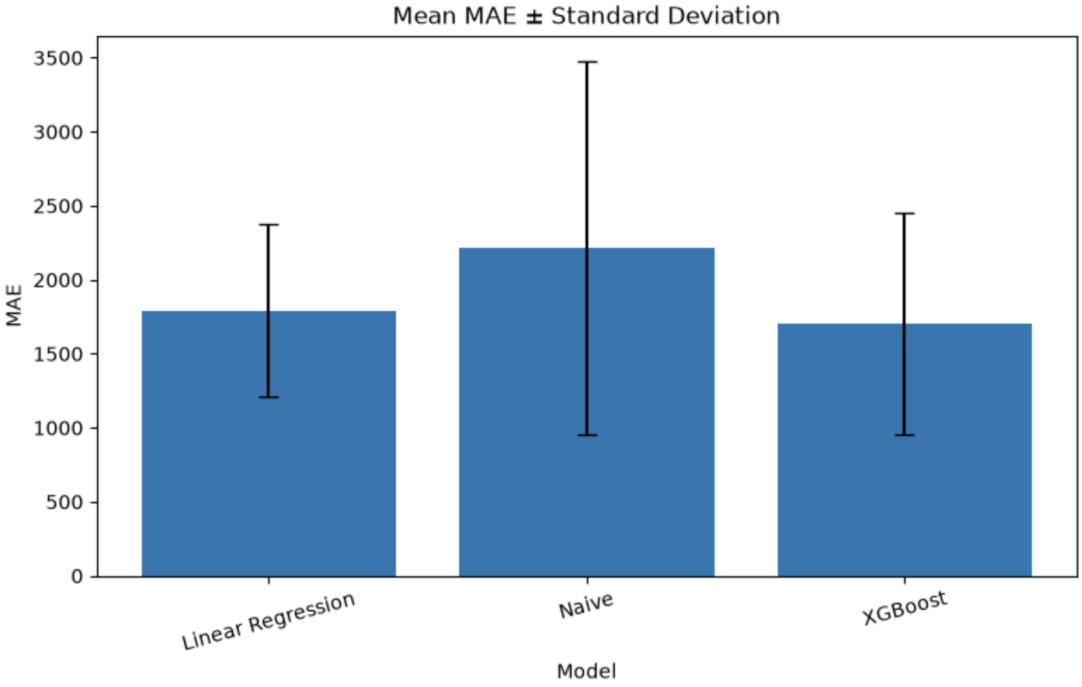


Figure 4. Mean MAE ± one standard deviation across six folds. Lower is better. Error bars show variability between test periods, not confidence intervals.

5.1 Fold-level results:

	fold	model	MAE	RMSE	MASE
0	1	Naive	1712.755157	3967.546193	1.019587
1	1	Linear Regression	1424.542898	3098.546736	0.848017
2	1	XGBoost	1434.915964	3241.465183	0.854192
3	2	Naive	4773.927795	16006.155477	2.821577
4	2	Linear Regression	2957.229690	8657.096638	1.747838
5	2	XGBoost	3214.453848	10108.773333	1.899867
6	3	Naive	1768.502439	3893.240128	0.754360
7	3	Linear Regression	1551.195584	3386.281703	0.661667
8	3	XGBoost	1430.758082	3100.632118	0.610294
9	4	Naive	1928.066170	4419.104854	0.860195
10	4	Linear Regression	1766.915556	3956.557391	0.788299
11	4	XGBoost	1603.560156	3791.313146	0.715419
12	5	Naive	1531.991467	3328.701469	0.700418
13	5	Linear Regression	1491.784367	2810.032624	0.682036
14	5	XGBoost	1242.399954	2538.906777	0.568018
15	6	Naive	1568.546174	3584.857763	0.742041
16	6	Linear Regression	1560.888157	3055.598736	0.738418
17	6	XGBoost	1295.973649	2713.697112	0.613093

6. Recommendation

It is recommend testing XGBoost together with the simple Naive model. We should not change the current ordering system immediately. XGBoost had an average MAE of 1,704, which is 23% lower than the Naive model's 2,214. Its mean MASE was 0.877. Also, it should be noticed that Linear Regression also performed better than the Naive model on average.

However, Linear Regression had a slightly lower mean RMSE (4,161) than XGBoost (4,249). This means Linear Regression performed better when considering larger errors. More over we can notice that the results also changed between test periods. Therefore, the company should test both models and check their performance before deciding which one to use in practical business case.

7.Suggested implementation:

• For a limited pilot, run Naive and XGBoost forecasts concurrently.

• Monitor MAE, RMSE, and MASE on a weekly basis, with findings broken down by department and store.

• We should check stock shortages, extra stock, and staff planning before saying the model saves money.

• When new sales data is available, we should update and test the models again. We should pay special attention to holiday weeks and departments where sales can suddenly increase.

8. Limitations and Reproducibility

- The data are from the past, so they may not show current sales or customer needs.
- The model predicts sales one week ahead. Longer forecasts may be less accurate.
- One model is used for all stores and departments, but their sales patterns may be different.
- The test covers only six periods, so future results may be different.
- The data may be missing information about promotions, stock availability, local events, and store operations.
- Accurate forecasts do not always mean more profit or better service. The company must test the model in real situations.

8.1. Reproducibility

Six chronological folds, ten test dates per fold, a one-date gap, the target Weekly_Sales, a lag-1 Naive baseline, Linear Regression, and XGBoost are all used in the analysis. There are eighteen rows in the results_per_fold.csv file (six folds × three models).Using the mean and sample standard deviation over six folds, summary statistics were recalculated from the provided per-fold CSV.

To reproduce the analysis, extract the project files, open the notebook in Jupyter Notebook or VS Code, install the required Python libraries, and run the notebook from top to bottom using a clean kernel. Confirm that the final results contain six folds and 18 rows before generating report tables and charts.

Note: AI is used for language correction. Proper language selection